

# US Patent & Trademark Office

## Patent Public Search | Text View

United States Patent Application Publication

20250287101

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

TAKAHASHI; Shigemi

### METHOD FOR CONTROLLING OPTICAL DEVICE

#### Abstract

An optical device includes an imager, an actuator to change the angle of the imager, and a gyro sensor to obtain information about a deviation angle of the imager. A method for controlling the optical device includes specifying a subject, defining an imaginary line connecting the centers of the imager and subject as a reference line, setting a threshold of the deviation angle of the imager in a direction intersecting the reference line, changing, with the actuator, the angle of the imager along the reference line, obtaining, with the gyro sensor, the information about the deviation angle, changing, with the actuator, the angle of the imager until the deviation angle falls below the threshold in response to the deviation angle exceeding the threshold, and continuing an operation of changing, with the actuator, the angle of the imager along the reference line after the deviation angle falls below the threshold.

**Inventors:** TAKAHASHI; Shigemi (Tokyo, JP)

**Applicant:** NIDEC PRECISION CORPORATION (Tokyo, JP)

**Family ID:** 96949794

**Appl. No.:** 19/071754

**Filed:** March 06, 2025

#### Foreign Application Priority Data

JP 2024-035355

Mar. 07, 2024

#### Publication Classification

**Int. Cl.:** H04N23/68 (20230101); H04N23/61 (20230101); H04N23/695 (20230101)

**U.S. Cl.:**

**CPC** H04N23/685 (20230101); H04N23/61 (20230101); H04N23/695 (20230101);

# Background/Summary

## RELATED APPLICATIONS

[0001] The present application claims priority to Japanese Application Number 2024-035355, filed Mar. 7, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

## BACKGROUND

### Technical Field

[0002] The disclosure relates to a method for controlling an optical device.

### Description of the Background

[0003] Patent Literature 1 describes a camera control device that controls a camera to pan, tilt, and zoom. The camera control device includes a face detector that detects the face of a presenter in an image captured with the camera, and a first control unit that controls the camera to pan, tilt, or zoom to allow the face to be detected when the face is not detected by the face detector.

## CITATION LIST

### Patent Literature

[0004] Patent Literature 1: Japanese Unexamined Patent Application Publication No. 2010-087613

## BRIEF SUMMARY

[0005] When the angle of an imager is changed toward a subject as in the structure in Patent Literature 1, the imager may receive vibrations. For example, an optical device may receive vibrations from, for example, an earthquake during imaging of a subject with the optical device placed on a desk, or may receive vibrations when the subject is walking and taking a selfie with the optical device. As described above, the optical device may receive vibrations in changing the angle of the imager toward the subject, possibly causing image blur.

[0006] One or more aspects of the disclosure are directed to a technique for reducing image blur that may occur with vibrations received by an optical device in changing the angle of an imager toward a subject.

[0007] A method according to an aspect of the disclosure is a method for controlling an optical device. The optical device includes an imager, a drive to change an angle of the imager, and an obtainer to obtain information about a deviation angle from a reference angle of the imager. The method includes specifying at least one subject, defining, in response to the at least one subject being specified, an imaginary line connecting a center of the imager and a center of the at least one subject as a reference line representing the reference angle, setting at least one threshold of the deviation angle from the reference angle of the imager in an intersecting direction intersecting with the reference line, starting an operation of changing, with the drive, the angle of the imager along the reference line, obtaining, with the obtainer, the information about the deviation angle in the intersecting direction, changing, with the drive, the angle of the imager until the deviation angle in the intersecting direction falls below the at least one threshold in response to the deviation angle in the intersecting direction exceeding the at least one threshold in changing, with the drive, the angle of the imager along the reference line, and continuing the operation of changing, with the drive, the angle of the imager along the reference line after the deviation angle in the intersecting direction falls below the at least one threshold.

[0008] The method according to the above aspect of the disclosure can reduce image blur that may occur with vibrations received by the optical device in changing the angle of the imager toward the subject.

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## Description

## BRIEF DESCRIPTION OF DRAWINGS

[0009] FIG. 1 is a diagram describing an optical device according to a first embodiment operated with a communication terminal.

[0010] FIG. 2 is a block diagram of the optical device and the communication terminal shown in FIG. 1, showing their internal structures.

[0011] FIG. 3 is a partial cross-sectional view of the optical device shown in FIG. 1, showing its internal structure.

[0012] FIG. 4 is a partial plan view of the optical device shown in FIG. 3, showing its internal structure.

[0013] FIG. 5 is a partial perspective view of the optical device shown in FIG. 3, showing its internal structure.

[0014] FIG. 6 is a diagram showing image patterns of a subject obtained by operating the optical device placed on a desk.

[0015] FIG. 7 is a table showing correction patterns for a deviation angle that can be set in the optical device shown in FIG. 1.

[0016] FIG. 8 is a graph showing, for example, a reference line used to correct the deviation angle of the optical device shown in FIG. 1.

[0017] FIG. 9 is a flowchart of the processes for correcting the deviation angle in the optical device shown in FIG. 1.

[0018] FIG. 10 is a graph showing correction of the deviation angle of the optical device shown in FIG. 1.

[0019] FIG. 11 is a diagram of an optical device according to a second embodiment capturing an image of the face of a subject who is seated and an image of the face of the subject who is standing.

[0020] FIG. 12 is a block diagram of a communication terminal in the second embodiment, showing its internal structure.

[0021] FIG. 13A is a graph showing reference lines and boundaries set for the optical device shown in FIG. 11 when the face position of the subject changes.

[0022] FIG. 13B is a graph showing, when the face position of the subject changes in the optical device shown in FIG. 11, the acceleration of a drive to change the angle of the imager changed at time points.

[0023] FIG. 14 is a flowchart of the processes for correcting the deviation angle in imaging performed with the subject tracked by the optical device shown in FIG. 11.

[0024] FIG. 15 is a diagram of the subject taking a selfie with an optical device according to a third embodiment.

[0025] FIG. 16 is a graph showing a reference line, boundaries, and limit lines set in the optical device shown in FIG. 15.

[0026] FIG. 17 is a flowchart of the processes for correcting the deviation angle in imaging performed with the subject tracked by the optical device shown in FIG. 15.

[0027] FIG. 18 is a diagram showing image patterns when the orientation of the imager is switched from a first subject to a second subject in an optical device according to a fourth embodiment.

[0028] FIG. 19 is a graph showing reference lines and boundaries set when the subject is switched to the second subject during imaging of the first subject in the optical device shown in FIG. 18.

[0029] FIG. 20 is a flowchart of the processes for correcting the deviation angle in imaging performed with the subject switched in the optical device shown in FIG. 18.

## DETAILED DESCRIPTION

[0030] Embodiments and modifications of the disclosure will now be described in detail with reference to the drawings. In the drawings used to describe the embodiments and the modifications, the same reference numerals denote the same or substantially the same components or elements. The components or elements described once will not basically be described repeatedly. Unless

otherwise specified, the terms such as first and second will be used simply to distinguish the components and will not represent a specific order or sequence.

#### Components in First Embodiment

[0031] FIG. 1 shows an optical device **10** according to a first embodiment. The optical device **10** includes a body case **11**, an imager **12**, and an actuator **30**. The body case **11** has an opening **11A**. The imager **12** is accommodated in the body case **11**. The imager **12** is partially visible from outside the body case **11** through the opening **11A**. The actuator **30** changes the angle of the imager **12**.

[0032] The optical device **10** operates by, for example, transmitting and receiving information to and from a communication terminal **100**. The communication terminal **100** is a mobile terminal with wireless communication capabilities operable by a user. The user may not be a subject. The communication terminal **100** may be a smartphone or a tablet. The communication between the communication terminal **100** and the optical device **10** is performed using, for example, Bluetooth (registered trademark). The communication may be performed using Wi-Fi (registered trademark).

[0033] For example, a first subject **S1**, a second subject **S2**, and a selection frame **FR** appear on a display panel **108** (described later) in the communication terminal **100**.

#### Communication Terminal

[0034] As shown in FIG. 2, the communication terminal **100** includes a controller **102**, the display panel **108**, and a communication interface (I/F) **109**. The controller **102** includes a central processing unit (CPU) **104** and a memory **106**. The memory **106** stores multiple programs for controlling the optical device **10**.

[0035] The controller **102** controls the operation of the optical device **10**. The multiple programs stored in the controller **102** include a program for changing the angle of the imager **12** (described later) and correcting the deviation angle of the imager **12**. The processes performed with the above program will be described later with reference to a flowchart (FIG. 9).

[0036] The display panel **108** functions as a touchscreen. On the display panel **108**, subjects including the user can appear or various items can be set through the operation by the user. The setting using the display panel **108** includes setting for user face authentication and setting for, for example, selecting subjects to be imaged. The communication I/F **109** wirelessly communicates with a communication I/F **26** (described later).

[0037] The selection frame **FR** (FIG. 1) that is quadrilateral appears on the display panel **108**. The user can move the selection frame **FR** to an intended position by touching the display panel **108** with a finger. For example, the selection frame **FR** is set around the face of the first subject **S1** (FIG. 1). The center of a target in the selection frame **FR** is at the center of an image obtained by the imager **12**. In other words, the actuator **30** operates to change the angle of the imager **12** and cause the center of a target specified with the selection frame **FR** to be at the center of an image.

#### Optical Device

[0038] As shown in FIG. 2, the optical device **10** includes the body case **11**, the imager **12**, a controller **20**, the actuator **30**, a detector **72**, a gyro sensor **78**, the communication I/F **26**, and a battery (not shown). The controller **20** includes a CPU **22** and a memory **24**. The memory **24** stores a program for transmitting and receiving signals to and from components of the optical device **10**.

[0039] The controller **20** causes the components of the optical device **10** to operate based on commands received from the communication terminal **100**. The controller **20** mediates transmission and reception of signals between the controller **102** and the components of the optical device **10**. The communication I/F **26** wirelessly communicates with the communication I/F **109**.

#### Imager

[0040] The imager **12** shown in FIG. 3 includes a lens **14** as an optical element and a body **16**. The body **16** includes, for example, an image sensor **18** that captures a subject image formed by the lens **14**. The imager **12** with this structure captures an image of a subject. The imager **12** has an optical axis **Z1**. The imager **12** is rotated in a pan direction and a tilt direction by the actuator **30**. The

imager **12** is fixed to a compartment **38** in a holder **36** (described later). Although not described in the present embodiment, the imager **12** may include a focuser including a lens and a driver.

#### Actuator

[0041] The actuator **30** shown in FIG. **3** is an example of a drive to change the angle of the imager **12** (FIG. **2**). The angle of the imager **12** specifically refers to the angle of the optical axis **Z1** in the pan direction and the tilt direction. The angle of the imager **12** is changed by changing the angle of the optical axis **Z1** in the pan direction and the tilt direction. In this manner, the actuator **30** changes the angle of the imager **12** in the pan direction and the tilt direction.

[0042] The arrows X, Y, and Z in the figures referred to in the examples below respectively indicate the X-direction, the Y-direction, and the Z-direction. For each of the X-direction, the Y-direction, and the Z-direction, the tip of the arrow points in one direction (positive direction) and the base of the arrow points in the other direction (negative direction). In the examples described below, the directions may be referred to as the positive X-direction, the negative X-direction, the positive Y-direction, the negative Y-direction, the positive Z-direction, and the negative Z-direction.

[0043] In the present embodiment, when the imager **12** is at an initial position before rotation (hereafter referred to as a reference position), the X-direction is the pan direction, the Y-direction is the tilt direction, and the Z-direction is an optical axis direction. The relationship between the X-, Y-, and Z-directions and the pan, tilt, and optical axis directions in the present embodiment is a mere example, and is not limitative. When the imager **12** is at the reference position, the X-direction, the Y-direction, and the Z-direction are perpendicular to one another. A point at which the X-axis, the Y-axis, and the Z-axis intersect with one another is defined as a reference point C.

[0044] The actuator **30** shown in FIG. **4** includes, for example, a case **32**, the holder **36**, a support assembly **40**, two first magnets **56**, two second magnets **58**, and a coil unit **62**. The actuator **30** further includes a flexible printed circuit (FPC) **68**, a driver integrated circuit (IC) **70**, a first sensor **74** and a second sensor **76**, and the gyro sensor **78** (FIG. **2**).

#### Case

[0045] As shown in FIG. **4**, the case **32** includes a bottom wall **33**, support walls **34** and **35**, and a side wall (not shown). The bottom wall **33** is a disk having a predetermined thickness in the Z-direction. The support walls **34** and **35** stand upright from the bottom wall **33** in the positive Z-direction and face each other in the X-direction. The holder **36** (described later) is accommodated in a space surrounded by the bottom wall **33** and the support walls **34** and **35**.

#### Holder

[0046] As shown in FIG. **5**, the holder **36** is spherical. In the present embodiment, being spherical refers to a spherical portion that can include a non-spherical portion such as a flat portion or a curved portion. The holder **36** has an opening **37** and the compartment **38** (FIG. **3**). The compartment **38** accommodates the imager **12** (FIG. **2**). The holder **36** includes the two first magnets **56**, two first yokes **57** (FIG. **3**), the two second magnets **58**, and two second yokes **59** (FIG. **3**).

[0047] As shown in FIG. **3**, the holder **36** has a spherical surface centered at the reference point C that is preset. In the present embodiment, for example, the rotation center of the holder **36** is at the reference point C. The rotation center of the holder **36** may not be at the reference point C. For the rotation direction of the holder **36**, the clockwise rotation in the figure is referred to as a rotation in the positive R-direction, and the counterclockwise rotation in the figure is referred to as a rotation in the negative R-direction.

#### Support Assembly

[0048] As shown in FIG. **4**, the support assembly **40** is located inside the case **32**. The support assembly **40** supports the holder **36** in a manner rotatable about the reference point C. Lines including the reference point C and extending in a direction intersecting with the radial direction of the holder **36** when the reference point C is viewed in the positive Z-direction are referred to as rotation axes CA. The support assembly **40** includes, for example, a frame **42**, four bases **44**, and

rotational shafts **46**. In the present embodiment, the rotation axes CA include a first rotation axis CX and a second rotation axis CY perpendicular to each other when the reference point C is viewed in the positive Z-direction. The first rotation axis CX extends in the X-direction. The second rotation axis CY extends in the Y-direction. Unless otherwise specified, the reference point C serves as a starting point from which the angle of the imager **12** is changed.

[0049] As shown in FIG. 5, the frame **42** has an annular shape centered at the reference point C (FIG. 4) when viewed in the Z-direction. The four bases **44** are arranged at equal intervals in the circumferential direction of the frame **42**. The four bases **44** are fastened to the frame **42** with multiple screws **49**.

[0050] As shown in FIG. 4, each rotational shaft **46** is located on the frame **42** in a manner rotatable about the corresponding rotation axis CA. More specifically, the rotational shafts **46** include a pair (two) of first shafts **47** rotatable about the first rotation axis CX, and a pair (two) of second shafts **48** supporting the holder **36** in a manner rotatable about the second rotation axis CY. As described above, the support assembly **40** includes the frame **42**, the pair of first shafts **47**, and the pair of second shafts **48**.

[0051] The pair of first shafts **47** and the pair of second shafts **48** are located on the four bases **44**. The first shafts **47** and the second shafts **48** are non-magnetic. In the present embodiment, being non-magnetic refers to having a relative permeability less than 1.5. Being ferromagnetic refers to having a relative permeability greater than or equal to 1.5.

[0052] The pair of first shafts **47** are supported by the support walls **34** and **35** of the case **32** in a rotatable manner. The pair of second shafts **48** support the holder **36** in a rotatable manner. In other words, the pair of first shafts **47** rotate relative to the support walls **34** and **35**. The pair of second shafts **48** rotate relative to the holder **36**. This allows the holder **36** to rotate about the first rotation axis CX in the pan direction and about the second rotation axis CY in the tilt direction. The position of the holder **36** at which the opening **37** in the holder **36** is open in the positive Z-direction is referred to as a reference position of the actuator **30**. The rotation angle of the holder **36** at the reference position is defined as 0°.

[0053] A first magnetic member **52** is fixed to the first shaft **47** in the positive X-direction. The first magnetic member **52** is annular. The direction of the magnetic force of the first magnetic member **52** changes as the first shaft **47** rotates. The magnetic force of the first magnetic member **52** is detected by the first sensor **74** (described later).

[0054] A second magnetic member **54** is fixed to the second shaft **48** in the positive Y-direction. The second magnetic member **54** is annular. The direction of the magnetic force of the second magnetic member **54** changes as the second shaft **48** rotates. The magnetic force of the second magnetic member **54** is detected by the second sensor **76** (described later).

#### First Magnet

[0055] As shown in FIG. 3, each first magnet **56** is fixed to the holder **36** with the corresponding first yoke **57** between them. The first magnet **56** is located on one end in the radial direction from the reference point C. The first magnet **56** has two N poles and one S pole between the two N poles.

#### Second Magnet

[0056] Each second magnet **58** is fixed to the holder **36** with the corresponding second yoke **59** between them. The second magnet **58** is located on the other end in the radial direction from the reference point C. The second magnet **58** has one N pole and one S pole. The first magnets **56** and the second magnets **58** are located asymmetric to each other with respect to the reference point C and have different lengths in the circumferential direction of the holder **36**.

[0057] The direction in which a first magnet **56** and a second magnet **58** in a first set are arranged is referred to as a K1-direction. The direction in which a first magnet **56** and a second magnet **58** in a second set are arranged is referred to as a K2-direction (FIG. 4). The K1-direction and the K2-direction are perpendicular to each other. The K1-direction and the K2-direction each intersect with

the X-direction and the Y-direction. The K1-direction and the K2-direction are each offset from the X-direction and the Y-direction by an angle of 45° when viewed in the Z-direction.

[0058] To distinguish the first magnet **56** and the second magnet **58** in the first set and the first magnet **56** and the second magnet **58** in the second set, the first magnet **56** and the second magnet **58** in the first set are referred to as a first magnet **56A** and a second magnet **58A**, and the first magnet **56** and the second magnet **58** in the second set are referred to as a first magnet **56B** and a second magnet **58B**. The first magnet **56A** and the second magnet **58A** are arranged in the K1-direction. The first magnet **56B** and the second magnet **58B** are arranged in the K2-direction.

#### Coil Unit

[0059] As shown in FIG. 4, the coil unit **62** is located on the support walls **34** and **35** of the actuator **30**. The coil unit **62** includes a first coil **63**, a second coil **64**, a third coil **65**, and a fourth coil **66**. The first coil **63** and the second coil **64** are spaced from each other in the K1-direction. The third coil **65** and the fourth coil **66** are spaced from each other in the K2-direction. The coil unit **62** faces the first magnets **56** and the second magnets **58** in the radial direction of the holder **36**. Thus, the coil unit **62** is energized to generate magnetic fields that act on the first magnets **56** and the second magnets **58**.

[0060] As shown in FIG. 3, a line including the reference point C and extending in the K1-direction is referred to as an imaginary line H. One S pole of the first magnet **56** extends in the negative Z-direction and the positive Z-direction across the imaginary line H in the R-direction. The second magnet **58** has a boundary surface (polarizing surface) between the N pole and the S pole on the imaginary line H. In the actuator **30**, the components are arranged in the K2-direction in the same manner as in the K1-direction. The components in the K2-direction will not be described.

#### Flexible Printed Circuit (FPC)

[0061] As shown in FIG. 4, the FPC **68** is placed from outside the case **32** into the case **32** and extends along the bottom wall **33**. The driver IC **70** (described later) is mounted on and connected to a part of the FPC **68**. The FPC **68** branches on the bottom wall **33** and is electrically connected to the coil unit **62**, the first sensor **74**, and the second sensor **76**.

#### Driver IC

[0062] The driver IC **70** shown in FIG. 4 drives the coil unit **62** to rotate the holder **36** about the first rotation axis CX and the second rotation axis CY. The rotation of the holder **36** will be described later. The driver IC **70** is electrically connected to the controller **20** (FIG. 2).

#### Detector

[0063] The detector **72** detects the angle (the pan angle and the tilt angle) of the imager **12**. Information about the angle of the imager **12** detected by the detector **72** is transmitted to the controller **20** (FIG. 2). The detector **72** includes the first sensor **74** and the second sensor **76**.

[0064] The first sensor **74** is a tunnel magneto resistance (TMR) sensor. The first sensor **74** is located on the support wall **34**. The first sensor **74** detects the rotational position of the holder **36** about the first rotation axis CX by detecting the direction of the magnetic force of the first magnetic member **52**.

[0065] The second sensor **76** is a TMR sensor. The second sensor **76** is located on the frame **42**. The second sensor **76** detects the rotational position of the holder **36** about the second rotation axis CY by detecting the direction of the magnetic force of the second magnetic member **54**.

Information about the rotational position of the holder **36** about the first rotation axis CX and information about the rotational position of the holder **36** about the second rotation axis CY are converted into information about the angle in the pan direction and the angle in the tilt direction.

#### Gyro Sensor

[0066] The gyro sensor **78** shown in FIG. 2 is an example of an obtainer to obtain information about the deviation angle of the imager **12** from its reference angle. The reference angle of the imager **12** refers to an angle of an imaginary line in the pan direction and an angle of the imaginary line in the tilt direction. The imaginary line is a line connecting the center of the imager **12** and the

center of a subject (a subject in a stationary state) that is an imaging target with the center of the imager **12** defined as an origin (starting point). In the examples described below, the imaginary line is referred to as a reference line M (FIG. **8**) representing a reference angle.

[0067] In the present embodiment, for example, the center of the imager **12** is at the reference point C (FIG. **3**). The center of the imager **12** refers to the center of the holder **36** (FIG. **3**) including the imager **12**. The center of the imager **12** may be set at the center position of the image sensor **18** or the position of the center of gravity of the image sensor **18**. The center of a subject refers to the center of a portion of the subject imaged by the imager **12**. For example, when the subject is a human and the face is imaged, the center of the subject refers to the center of the face. When the entire body is imaged, the center of the subject refers to the center of the body.

[0068] The gyro sensor **78** detects the angular velocity of the actuator **30** (optical device **10**). The angular velocity detected by the gyro sensor **78** is integrated to calculate deviation angles (angle deviations) in the pan direction and the tilt direction. The obtained angular velocity may be integrated by the gyro sensor **78** or by the CPU **22**.

[0069] A deviation angle obtained by the gyro sensor **78** can be determined as the deviation angle of the imager **12** based on disturbance in an intersecting direction intersecting with the reference line M (described later) in changing the angle of the imager **12**. The intersecting direction is not limited to a direction that forms a right angle with the reference line M, and may be a direction intersecting with the reference line M at an angle other than a right angle.

[0070] For example, the deviation angle (with components in the pan direction and the tilt direction) of the imager **12** from the reference line M may be a deviation angle  $\theta\alpha$  (°) during driving of the holder **36** (FIG. **3**) with the actuator **30** to cause the imager **12** to be at an angle on the reference line M. The deviation angle (the deviation angle based on disturbance such as vibrations) obtained by the gyro sensor **78** may be a deviation angle  $\theta\beta$  (°). In this case, the actual deviation angle  $d\theta$  of the imager **12** from the reference line M is expressed as  $d\theta = \theta\alpha + \theta\beta$ . The actuator **30** then changes, based on a command input through the controller **20**, the angle of the imager **12** toward the reference line M, possibly decreasing the deviation angle  $d\theta$  to 0°. The deviation angle  $\theta\alpha$  and the deviation angle  $\theta\beta$  are not shown in the figures.

#### Rotation of Holder

[0071] In the actuator **30** shown in FIG. **4**, in response to the coil unit **62** being energized, thrust may act on the first magnet **56A** and the second magnet **58B** in the negative Z-direction and on the first magnet **56B** and the second magnet **58A** in the positive Z-direction. In this case, the holder **36** rotates in one direction about the second rotation axis CY. When the thrust acts in the opposite directions, the holder **36** rotates in the other direction about the second rotation axis CY. In other words, the imager **12** and the holder **36** rotate in the tilt direction.

[0072] In response to the coil unit **62** being energized, thrust may act on the first magnet **56A** and the first magnet **56B** in the positive Z-direction and on the second magnet **58A** and the second magnet **58B** in the negative Z-direction. In this case, the holder **36** rotates in one direction about the first rotation axis CX. When the thrust acts in the opposite directions, the holder **36** rotates in the other direction about the first rotation axis CX. In other words, the imager **12** and the holder **36** rotate in the pan direction. In this manner, the actuator **30** changes the angle of the imager **12** in the pan direction and the tilt direction. The angle of the imager **12** has components in both the pan direction and the tilt direction perpendicular to each other.

#### Control of Optical Device Using Communication Terminal

[0073] As shown in FIG. **6**, the optical device **10** is placed on the top of a desk D. The optical device **10** is supported on a support plate (not shown). For example, the first subject S1 and the second subject S2 are in the range of an angle of view VA of the imager **12**. The imager **12** is oriented at the reference angle that is preset.

[0074] FIG. **7** collectively shows control patterns of the optical device **10** (FIG. **2**) in a table. The control patterns of the optical device **10** are each determined by a combination of an imaging mode

that can be set in the communication terminal **100** (FIG. 2) and the state of the optical device **10** in use. The control patterns are, for example, a pattern A, a pattern B, and a pattern C. The pattern A is a control pattern used when a normal mode or a target input mode (described later) is selected with the optical device **10** placed on the desk D (FIG. 6).

[0075] The pattern B is a control pattern used when a target tracking mode (described later) is selected with the optical device **10** placed on the desk D. The pattern C is a control pattern used when the target tracking mode is selected with the optical device **10** held by hand. For example, the normal mode or the target input mode is not selected with the optical device **10** held by hand. In the first embodiment, the control pattern is the pattern A. The pattern B and the pattern C will be described in the second and subsequent embodiments.

#### Reference Line and Allowable Range

[0076] FIG. 8 shows the reference line M used when the angle of the imager **12** (FIG. 3) is changed from an initial angle  $d0$  to a target angle  $d4$ . The angle of the imager **12** corresponds to the orientation of the imager **12** determined by the angle of the imager **12** in the tilt direction (tilt angle) and the angle of the imager **12** in the pan direction (pan angle). The initial angle  $d0$  refers to a tilt angle  $\theta A$  (°) and a pan angle  $\theta 0$  (°) when the imager **12** has an orientation that serves as a reference before the orientation of the imager **12** is changed. The target angle  $d4$  refers to a tilt angle  $\theta B$  (°) and a pan angle  $\theta 2$  (°) when the imager **12** reaches an orientation at which the imager **12** can capture an image of a specified subject.

[0077] The reference line M is an imaginary line connecting, in response to a subject being specified, the center of the imager **12** and the center of the subject. The reference line M is the shortest straight line connecting the coordinates of the initial angle  $d0$  and the coordinates of the target angle  $d4$ . The reference line M represents a reference angle that serves as a reference for control in changing the angle of the imager **12**. In other words, the reference line M represents changes in the tilt angle and changes in the pan angle in controlling the operation of the actuator **30** (FIG. 3).

[0078] The reference line M indicates angles  $d1$ ,  $d2$ , and  $d3$  each representing the angle of the imager **12** during change of the angle of the imager **12** from the initial angle  $d0$  to the target angle  $d4$ . The operation of the actuator **30** (FIG. 3) is controlled along the reference line M, thus changing the tilt angle and the pan angle of the imager **12**. This allows the imager **12** to be oriented toward the subject.

[0079] When the actuator **30** operates based on the reference line M, the angle of the imager **12** changes in the order of the initial angle  $d0$ , the angle  $d1$ , the angle  $d2$ , the angle  $d3$ , and the target angle  $d4$  over time. In FIG. 8, the intersecting direction intersecting with the reference line M is indicated by an arrow Q. The amount of deviation from the reference line M in the intersecting direction represents the amount of deviation of the angle of the imager **12**, or in other words, the deviation angle of the imager **12**. The angle between the intersecting direction and the reference line M is not limited to a right angle and may be another angle.

[0080] The deviation angle of the imager **12** from the reference line M is denoted with  $d\theta$  (°). In the present embodiment, thresholds (°) of the deviation angle  $d\theta$  including measurement errors are predetermined. For example, the same threshold of the deviation angle  $d\theta$  is set for each angle from the initial angle  $d0$  to the target angle  $d4$ . In FIG. 8, a circle AR having a radius being the threshold of the deviation angle  $d\theta$  is indicated. The deviation angle  $d\theta$  being within the circle AR refers to the deviation angle  $d\theta$  being within an allowable range.

[0081] A range through which the circle AR passes along the reference line M (a range hatched with diagonal lines) is defined as an allowable range A1 of the deviation angle  $d\theta$  of the imager **12**. The allowable range A1 indicates a range of allowable deviation angles from the reference line M for the tilt angle and the pan angle. A line segment in the allowable range A1 indicating the lower limit is defined as a boundary BL. A line segment in the allowable range A1 indicating the upper limit is defined as a boundary UL. In other words, the allowable range A1 is a range between the

boundary BL and the boundary UL.

## Effects of Components in First Embodiment

[0082] A method for controlling the optical device **10** will now be described with reference to the flowchart shown in FIG. **9**. For the components of the optical device **10** (FIG. **2**) and the communication terminal **100** (FIG. **2**), FIGS. **1** to **5** are to be referred to, and the figure numbers are not individually referred to. The processes shown in FIG. **9** are performed by the CPU **104** that loads and executes a program (not shown) after reading the program from the memory **106**. For example, the program starts in response to a subject being set by the user operating the communication terminal **100**. The angle of the imager **12** is then changed to cause the center of the set (selected) subject to be at the center of the display panel **108**.

[0083] In the overall processing in the flowchart, the angle of the imager **12** is set as the initial angle **d0**, and then the target angle **d4** is set. When any deviation angle occurs in the angle of the imager **12** in changing the angle of the imager **12** from the initial angle **d0** to the target angle **d4** after the operation of the actuator **30** is started, correction of the deviation angle is performed. The deviation angle (image blur) of the imager **12** may occur with, for example, earthquake vibrations received by the optical device **10** or camera shake from the user holding the optical device **10** by hand. In response to the angle of the imager **12** falling within the allowable range of the target angle **d4** (being less than the set threshold), the program ends.

[0084] In FIG. **9**, the actuator **30** is abbreviated as ACT. In the flowcharts in the figures other than FIG. **9** as well, the actuator **30** is abbreviated as ACT. A target refers to a subject to be imaged. The actuator **30** changes the angle of the imager **12** within a physically limited range. The upper limit value of the target angle **d4** is thus set to, for example, an angle of 37.5°. The upper limit value of the target angle **d4** may be set to an angle other than 37.5°. An ACT angle refers to the tilt angle and the pan angle of the imager **12**.

[0085] In step **S10**, the CPU **104** transmits command information to the driver IC **70** through the controller **20** to turn on a servo in the actuator **30**. The processing then advances to step **S12**.

[0086] In step **S12**, the CPU **104** detects the angle of the actuator **30** (the pan angle and the tilt angle of the imager **12**) based on detection information from the detector **72**. The processing then advances to step **S14**.

[0087] In step **S14**, the CPU **104** determines whether the difference between the detected angle of the actuator **30** and the initial angle **d0** is less than or equal to 1°. When the difference between the angle of the actuator **30** and the initial angle **d0** is less than or equal to 1° (Yes in **S14**), the CPU **104** determines that the angle of the actuator **30** is the initial angle **d0** and advances the processing to step **S16**. When the difference between the angle of the actuator **30** and the initial angle **d0** is greater than 1° (No in **S14**), the processing advances to step **S18**.

[0088] In step **S16**, the CPU **104** starts detecting the deviation angle **dθ** from the reference line **M** using the gyro sensor **78**. The processing then advances to step **S20**.

[0089] In step **S18**, the CPU **104** causes the actuator **30** to operate to cause the difference between the angle of the actuator **30** and the initial angle **dθ** to be less than or equal to 1°. More specifically, the CPU **104** changes the states of the first coil **63**, the second coil **64**, the third coil **65**, and the fourth coil **66** energized by the driver IC **70** to change the angle of the imager **12**. The processing then advances to step **S12**.

[0090] In step **S20**, the CPU **104** specifies a target based on information specified (set) on the display panel **108** by a user (target **S**). In this example, the first subject **S1** is specified, for example. The processing then advances to step **S22**. Step **S20** is an example of specifying a subject.

[0091] In step **S22**, the CPU **104** obtains information about the target angle **d4**. For example, the CPU **104** obtains information about the orientation (the tilt angle and the pan angle) of the first subject **S1** relative to the center of the imager **12** based on the positional information of the first subject **S1** on the display panel **108**. More specifically, a data table for the relationship between the position of a target appearing on the display panel **108** and the orientation of the first subject **S1** is

stored into the memory **106**, and the CPU **104** compares the positional information of the target with the information in the data table to obtain the orientation of the first subject **S1**. The processing then advances to step **S24**.

[0092] In step **S24**, the CPU **104** determines whether the target angle **d4** is less than or equal to  $37.5^\circ$  that is an upper limit angle. When the target angle **d4** is less than or equal to  $37.5^\circ$  (Yes in **S24**), the processing advances to step **S26**. When the target angle **d4** is greater than  $37.5^\circ$  (No in **S24**), the processing advances to step **S28**. The target angle **d4** being less than or equal to  $37.5^\circ$  refers to both the tilt angle and the pan angle being less than or equal to  $37.5^\circ$ .

[0093] In step **S26**, the CPU **104** sets the target angle **d4** in the memory **24** in the optical device **10**. The processing then advances to step **S30**.

[0094] In step **S28**, the CPU **104** sets the target angle **d4** as  $37.5^\circ$  in the memory **24** in the optical device **10**. The processing then advances to step **S30**.

[0095] In step **S30**, the CPU **104** defines the reference line **M** associated with the tilt angle and the pan angle based on information about the initial angle **d0** and the information about the target angle **d4**. The processing then advances to step **S32**. Step **S30** is an example of defining, in response to the first subject **S1** being specified, an imaginary line connecting the center of the imager **12** and the center of the first subject **S1** as the reference line **M** representing the reference angle.

[0096] In step **S32**, the CPU **104** generates a data table of the allowable range **A1** (thresholds) of the deviation angle **d0**. More specifically, the CPU **104** creates a table of combinations of the values of the tilt angle and the values of the pan angle, and then determines, for the created table, combinations of thresholds corresponding to the boundaries **BL** and **UL**. The CPU **104** can thus determine whether the angle of the imager **12** is within the allowable range **A1** based on the thresholds. The processing then advances to step **S34**. Step **S32** is an example of setting a threshold of the deviation angle **d0** from the reference angle of the imager **12** in the intersecting direction intersecting with the reference line **M**.

[0097] In step **S34**, the CPU **104** starts the operation of the actuator **30**. More specifically, the CPU **104** causes the actuator **30** to operate (a current to flow through the coils) to change the angle of the imager **12** from the initial angle **d0** to the target angle **d4** along the reference line **M**. Step **S34** is an example of starting an operation of changing, with the actuator **30**, the angle of the imager **12** along the reference line **M**. The processing then advances to step **S36**.

[0098] In step **S36**, the CPU **104** determines whether the difference between the tilt angle detected by the detector **72** and the tilt angle of the target angle **d4** and the difference between the pan angle detected by the detector **72** and the pan angle of the target angle **d4** (collectively referred to as an angular difference) are each greater than  $1^\circ$ . When the angular difference is greater than  $1^\circ$  (Yes in **S36**), the CPU **104** determines that the imager **12** has yet to reach the target angle **d4** and advances the processing to step **S38**. When the angular difference is less than or equal to  $1^\circ$  (No in **S36**), the CPU **104** determines that the imager **12** has reached the target angle **d4** and ends the program.

[0099] In step **S38**, the CPU **104** obtains the information about the tilt angle and the information about the pan angle (deviation angle **d01**) detected by the detector **72**, and obtains, from the gyro sensor **78**, information about the tilt angle and the pan angle (deviation angle **d02**). The total deviation angle **d0** is expressed as  $d0 = d01 + d02$ . The deviation angle **d01** and the deviation angle **d02** are not shown in the figures. Step **S38** is an example of obtaining, with the gyro sensor **78**, the information about the deviation angle **d02** in the intersecting direction. The processing then advances to step **S40**.

[0100] In step **S40**, the CPU **104** determines whether the deviation angle **d0** is greater than the threshold based on information about the total deviation angle **d0** obtained in step **S38**. More specifically, the CPU **104** compares the value of the total deviation angle **d0** with the corresponding threshold in the generated table. When the total deviation angle **d0** is greater than the threshold (Yes in **S40**), the processing advances to step **S42**. When the total deviation angle **d0** is less than or equal to the threshold (No in **S40**), the processing advances to step **S34**.

[0101] In step S42, the CPU 104 sets deviation angle reduction information in the driver IC 70 through the controller 20. More specifically, the CPU 104 sets, in the driver IC 70, the amount of movement of the actuator 30 (the amount of change in the angle of the imager 12) to cause the total deviation angle  $d\theta$  to be less than or equal to the threshold. The processing then advances to step S34. Steps S40 and S42 are examples of changing, with the actuator 30, the angle of the imager 12 until the deviation angle  $d\theta$  in the intersecting direction falls below the threshold in response to the deviation angle  $d\theta$  in the intersecting direction exceeding the threshold in changing, with the actuator 30, the angle of the imager 12 along the reference line M. Further, step S34 is an example of continuing the operation of changing, with the actuator 30, the angle of the imager 12 along the reference line M after the deviation angle  $d\theta$  in the intersecting direction falls below the threshold.

[0102] FIG. 10 shows the deviation angle  $d\theta$  exceeding the allowable range A1 (the thresholds: the boundary BL) during change of the orientation of the imager 12 (FIG. 2) to the target angle  $d4$  based on the reference line M. In FIG. 10, the actual changes in the angle of the imager 12 are indicated by an imaginary line G. The angle of the imager 12 is within the allowable range A1 along the reference line M until the imager 12 reaches an angle  $dA$ . However, when the optical device 10 (FIG. 2) receives, for example, vibrations with the imager 12 being at the angle  $dA$ , the deviation angle  $d\theta$  increases, causing the angle of the imager 12 to be an angle  $dB$  exceeding the allowable range A1.

[0103] In response to the deviation angle  $d\theta$  exceeding the allowable range A1, the actuator 30 (FIG. 2) changes (corrects) the tilt angle and the pan angle of the imager 12 to cause the deviation angle  $d\theta$  of the imager 12 to fall within the allowable range A1. The angle of the imager 12 is thus within the allowable range A1 again as indicated by the imaginary line G, reducing the deviation angle of the imager 12 from the reference line M. The actuator 30 then continues the operation of changing the angle of the imager 12 to the target angle  $d4$ .

[0104] As described above, the method for controlling the optical device 10 according to the first embodiment defines the reference line M connecting the center of the imager 12 and the center of the first subject S1 (the center of the selection frame FR) in response to the first subject S1 being specified on the display panel 108. Further, the threshold of the deviation angle  $d\theta$  of the imager 12 in the intersecting direction intersecting with the reference line M and the allowable range A1 are set. The actuator 30 then changes the angle of the imager 12 (the direction of the optical axis Z1) based on the reference line M. When the deviation angle  $d\theta$  obtained from the detector 72 and the gyro sensor 78 is within the allowable range A1, or in other words, when small vibrations occur in the optical device 10, the operation of the actuator 30 is continued and the angle of the imager 12 reaches the target angle  $d4$ . The imager 12 is thus oriented toward the first subject S1. An image of the first subject S1 thus appears at a center portion of the display panel 108 when imaging is performed with the imager 12.

[0105] When the deviation angle  $d\theta$  obtained from the detector 72 and the gyro sensor 78 exceeds the threshold and thus the angle of the imager 12 exceeds the allowable range A1, or in other words, when large vibrations occur in the optical device 10, the angle of the imager 12 is changed to reduce the deviation angle  $d\theta$ . The angle of the imager 12 then continues to be changed based on the reference line M. In other words, the actuator 30 changes the angle of the imager 12 until the angle of the imager 12 falls within the allowable range A1. Thus, under large vibrations in the optical device 10, the actuator 30 can cause the angle of the imager 12 to reach the target angle  $d4$  while reducing a deviation that affects the image. An image of the first subject S1 thus appears at the center portion of the display panel 108 when imaging is performed with the imager 12.

[0106] As described above, when vibrations occur in the optical device 10 in which imaging is performed with the imager 12 oriented toward the first subject S1, the imager 12 can be oriented toward the first subject S1 independently of the vibration level. This can reduce image blur that may occur with vibrations received by the optical device 10 in changing the angle of the imager 12 toward the first subject S1.

[0107] The method for controlling the optical device **10** according to the first embodiment causes the actuator **30** to drive the imager **12** to an angle (orientation) having a component in the pan direction and a component in the tilt direction, and thus can change the angle of the imager **12** in the pan direction and the tilt direction.

#### Components in Second Embodiment

[0108] FIG. **11** shows an optical device **10** according to a second embodiment with the imager **12** changing its angle by following the motion of the subject S. The second embodiment differs from the first embodiment in the method for controlling the optical device **10**. The same components as in the first embodiment are denoted with the same reference numerals and will not be described.

[0109] The subject S is seated on a chair CH. The optical device **10** is placed on the desk D. A face SF of the subject S faces the imager **12**. A range B1 represents an imaging range in which the imager **12** can capture an image of the entire face SF with the subject S seated on the chair CH. A range B2 represents an imaging range in which the imager **12** can capture an image of the entire face SF with the subject S standing up from the chair CH.

[0110] The optical device **10** operates by transmitting and receiving information to and from a communication terminal **110**. The angle of the imager **12** in the optical device **10** is changed by, for example, the subject S operating the communication terminal **110**. In the present embodiment, the face SF is identified when the subject S is seated. When the subject S stands up, the imager **12** tracks the face SF as the face SF of the subject S moves.

#### Communication Terminal

[0111] As shown in FIG. **12**, the communication terminal **110** includes a controller **112**, the display panel **108**, and the communication I/F **109**. The controller **112** includes the CPU **104** and the memory **106**. Further, the controller **112** includes an identifier **114** and a movement detector **116** as functional components. In the controller **112**, the CPU **104** executes a program stored in the memory **106** to implement the functions of the identifier **114** and the movement detector **116**.

[0112] The controller **112** may include separate hardware modules as the identifier **114** and the movement detector **116**. The processes performed with the program in the controller **112** will be described later with reference to a flowchart (FIG. **14**).

#### Identifier

[0113] The controller **112** obtains, using a known face recognition technique, information about the face SF of the subject S based on an image captured with the imager **12** (FIG. **11**) and stores the information into the identifier **114**. The identifier **114** compares the face SF (FIG. **11**) specified by the subject S operating the display panel **108** with the prestored information about the face SF to identify the face SF specified by the subject S as the subject S or another person.

#### Movement Detector

[0114] The movement detector **116** detects movement of the identified subject S from a first position to a second position. More specifically, the movement detector **116** determines whether the center position of the face SF identified by the identifier **114** changes over time in the image to detect any movement of the subject S from the first position to the second position.

#### Reference Line and Allowable Range

[0115] FIG. **13A** shows a reference line M1 used when the angle of the imager **12** (FIG. **11**) is changed from the initial angle  $d\theta$  to the target angle  $d4$  and a reference line M2 used when the angle of the imager **12** is changed from the initial angle  $d\theta$  to a target angle  $d5$ . The reference line M2 is an example of a new reference line.

[0116] The target angle  $d5$  refers to a tilt angle  $\theta C$  (°) and a pan angle  $\theta 2$  (°). The pan angle for both the target angles  $d4$  and  $d5$  is the pan angle  $\theta 2$ . In other words, when changing the angle of the imager **12** from the target angle  $d4$  to the target angle  $d5$  as the face SF (FIG. **11**) moves, the actuator **30** (FIG. **2**) changes the tilt angle from  $\theta B$  to  $\theta C$  while maintaining the pan angle at  $\theta 2$ .

[0117] In FIG. **13A**, an intersecting direction intersecting with the reference line M1 is indicated by an arrow Q1. A direction intersecting with the reference line M2 is indicated by an arrow Q2. In

FIG. 13A, in response to the subject S (FIG. 11) standing up, the target angle is changed from d4 to d5 before the imager 12 (FIG. 11) reaches the target angle d4. For example, the reference line is switched from M1 to M2 from the time point at which the imager 12 reaches an angle dC on the reference line M1 to the time point at which the imager 12 reaches an angle dE on the reference line M2. The reference for the angle of the imager 12 in switching the reference line from M1 to M2 is indicated by a reference line M3. The angle of the imager 12 is changed along the reference line M2 and to the target angle d5 after the reference line is switched from M3 to M2 at the angle dE.

[0118] At the initial angle d0, an allowable range of a deviation angle (an example of an imager threshold) is indicated by the circle AR. At each of the target angles d4 and d5, the allowable range of the deviation angle (an example of a subject threshold) is indicated by a circle BR. The circle BR has a smaller radius than the circle AR. In other words, thresholds include the subject threshold and the imager threshold set for an area closer to the imager 12 than the subject threshold. The subject threshold is less than the imager threshold.

[0119] The lower limits of the thresholds for the reference line M1 are indicated by a boundary BL1. The upper limits of the thresholds are indicated by a boundary UL1. The range between the boundary BL1 and the boundary UL1 is defined as an allowable range A2 of the deviation angle. Similarly, the lower limits of the thresholds for the reference line M2 are indicated by a boundary BL2. The upper limits of the thresholds are indicated by a boundary UL2. The range between the boundary BL2 and the boundary UL2 is defined as an allowable range A3 of the deviation angle. For example, the allowable ranges A2 and A3 are both set to continuously narrow as the angle of the imager 12 is closer to the target angles d4 and d5. Boundaries corresponding to the reference line M3 are defined but not shown in the figure.

#### Acceleration Setting

[0120] FIG. 13B is a graph GC showing changes in an ACT velocity (m/s) at time points (s). The ACT velocity refers to the velocity at which the actuator 30 (FIG. 11) operates and corresponds to the velocity at which the angle of the imager 12 (FIG. 11) changes. The change of the ACT velocity with time is defined as an acceleration AC (m/s.sup.2) of the imager 12. A data table of the graph GC is prestored in the memory 106 (FIG. 12).

[0121] In the graph GC, the ACT velocity is V0 (=0) at a time point t0 at which the actuator 30 starts operating. The ACT velocity increases from V0 to V1 from the time point t0 to a time point t1. For example, the ACT velocity is maintained at V1 from the time point t1 to a time point t2. The ACT velocity decreases from V1 to V0 from the time point t2 to a time point t3. For example, the time from the time point t0 to the time point t1 is shorter than the time from the time point t2 to the time point t3.

[0122] The data table of the graph GC is set to satisfy  $\alpha1 > \alpha2$ , where the acceleration AC from the time point t0 to the time point t1 is  $\alpha1$ , the acceleration AC from the time point t1 to the time point t2 is zero (a constant velocity), and the acceleration AC from the time point t2 to the time point t3 is  $\alpha2$ . At the time point t3, the angle of the imager 12 reaches the target angle d5 (FIG. 13A). As described above, the acceleration AC of the actuator 30 changing the angle of the imager 12 is set smaller when the angle of the imager 12 is closer to the target angles d4 and d5 (closer to the reference lines M1 and M2).

#### Effects of Components in Second Embodiment

[0123] A method for controlling the optical device 10 according to the second embodiment will now be described with reference to the flowchart shown in FIG. 14. For the components of the optical device 10 (FIG. 11) and the communication terminal 100 (FIG. 12), FIGS. 1 to 5 and FIGS. 11 to 13B are to be referred to, and the figure numbers are not individually referred to. The same effects as in the first embodiment will not be described.

[0124] The processes shown in FIG. 14 are performed by the CPU 104 that loads and executes a program (not shown) after reading the program from the memory 106. For example, the program

starts in response to the subject S operating the communication terminal **110** to set (perform face authentication on) the face SF of the subject S. The angle of the imager **12** is then changed to cause the set face SF to be at the center portion of the display panel **108**.

[0125] The same steps as in the first embodiment are denoted with the same reference numerals and will not be described. More specifically, unlike in the flowchart (FIG. 9) in the first embodiment, step **S21** is set in place of step **S20**, and step **S44** is added.

[0126] In step **S21** subsequent to step **S16**, the CPU **104** performs, with the identifier **114**, face authentication on the face SF of the subject S specified by the subject S on the display panel **108**. In this example, the face SF of the subject S is identified based on the result of face authentication performed with the identifier **114**. In this manner, the subject S as a tracking target is specified in step **S21**. The processing then advances to step **S22**. Step **S21** is included in an example of identifying the subject S with the identifier **114** in response to the subject S being specified.

[0127] In step **S22**, when the specified subject S is seated on the chair CH, the CPU **104** obtains information about the target angle **d4**. When the specified subject S is standing, the CPU **104** obtains information about the target angle **d5**. The processing then advances to step **S24**.

[0128] In step **S26**, when the specified subject S is seated on the chair CH, the CPU **104** sets the target angle **d4**. When the specified subject S is standing, the CPU **104** sets the target angle **d5**. The processing then advances to step **S30**.

[0129] In step **S30**, when the target angle **d4** is set, the CPU **104** defines the reference line **M1** associated with the tilt angle and the pan angle based on information about the initial angle **d0** and the information about the target angle **d4**. Step **S30** is an example of defining, in response to the subject S being at a first position (a seated position: the target angle **d4**), an imaginary line connecting the center of the imager **12** and the first position as the reference line **M1** before the movement.

[0130] In step **S30**, when the target angle **d5** is set, the CPU **104** defines the reference line **M2** associated with the tilt angle and the pan angle based on the information about the initial angle **d0** and the information about the target angle **d5**. In other words, step **S30** is also an example of defining, in response to the movement detector **116** detecting movement of the subject S to a second position (a standing position: the target angle **d5**), an imaginary line connecting the center of the imager **12** and the second position as the new reference line. The processing then advances to step **S32**.

[0131] In step **S32**, for the target angle **d4**, the CPU **104** generates a data table of the allowable range **A2** (thresholds) of the deviation angle **d0**. For the target angle **d5**, the CPU **104** generates a data table of the new allowable range **A3** (new thresholds) of the deviation angle **d0**. Step **S32** is an example of setting, in response to the subject S being at the second position, a new threshold corresponding to the new reference line (reference line **M2**). The processing then advances to step **S34**.

[0132] In step **S34**, the CPU **104** starts the operation of the actuator **30**. More specifically, the CPU **104** causes the actuator **30** to operate to change the angle of the imager **12** from the initial angle **d0** to the target angle **d4** along the reference line **M1**. The CPU **104** also causes the actuator **30** to operate to change the angle of the imager **12** from the initial angle **d0** to the target angle **d5** along the reference line **M2**. Step **S34** is an example of changing the angle of the imager **12** along the new reference line. The processing then advances to step **S36**.

[0133] In step **S36**, when the difference between the target angle **d4** and the current angle (orientation) of the imager **12** is less than or equal to  $1^{\circ}$  (No in **S36**), the CPU **104** advances the processing to step **S44**. When the angular difference is greater than  $1^{\circ}$  (Yes in **S36**), the CPU **104** determines that the imager **12** has yet to reach the target angle **d4** and advances the processing to step **S38**.

[0134] Steps **S40** and **S42** are examples of changing, with the actuator **30**, the angle of the imager **12** until the deviation angle falls below the new thresholds in response to the deviation angle

exceeding the new thresholds in changing, with the actuator **30**, the angle of the imager **12** and continuing an operation of changing, with the actuator **30**, the angle of the imager **12** along the new reference line after the deviation angle falls below the new thresholds.

[0135] In step **S44**, the CPU **104** determines whether the position of the subject **S** (target angle) detected by the movement detector **116** is the same as the target angle obtained in step **S22**. In other words, the CPU **104** determines whether the position of the face **SF** in the image captured by the imager **12** is the same as its position at the time when step **S22** is performed.

[0136] When determining that the position of the face **SF** is the same as its position at the time when step **S22** is performed (Yes in **S44**), the CPU **104** determines that the subject **S** as a tracking target has not moved, and ends the program. When determining that the position of the face **SF** is different from its position at the time when step **S22** is performed (No in **S44**), the CPU **104** determines that the subject **S** as a tracking target is moving. The CPU **104** advances the processing to step **S22** and starts tracking the subject **S**. In the second embodiment, as described above, in response to the position of the face **SF** of the subject **S** changing from its position at which the subject **S** is seated to its position at which the subject **S** is standing, the target angle is changed from **d4** to **d5** and the angle of the imager **12** is also changed.

[0137] As described above, the method for controlling the optical device **10** according to the second embodiment causes the identifier **114** to perform face authentication on the face **SF** of the specified subject **S** to identify the subject **S**. When the identified subject **S** has moved, the movement detector **116** detects the position to which the subject **S** has moved based on the difference in the position of the face **SF**. In response to the reference line **M1** and the allowable range **A2** being respectively replaced with (set to) the new reference line **M2** and the allowable range **A3**, the angle of the imager **12** is changed. When the deviation angle of the imager **12** changes due to vibrations of the optical device **10**, correction is performed based on the deviation angle in the intersecting direction from the reference line **M2**, thus reducing blur in an image obtained by the imager **12**. In other words, this method can change the direction of the imager **12** to follow the subject **S**, and can also reduce image blur.

[0138] With the method for controlling the optical device **10** according to the second embodiment, the subject threshold is less than the imager threshold. In other words, the allowable range **A3** is smaller as the angle of the imager **12** is closer to the angle at which the imager **12** faces the subject **S**. This reduces a deviation between the imaging position of the face **SF** of the subject **S** when the imager **12** reaches the target angle **d5** on the reference line **M2** and the position of the face **SF** in an obtained actual image.

[0139] With the method for controlling the optical device **10** according to the second embodiment, the acceleration **AC** of the actuator **30** changing the angle of the imager **12** is smaller when the angle of the imager **12** is closer to the reference line **M1** or the reference line **M2**. This reduces the likelihood that the imager **12** moves unintendedly under its inertial force when the imager **12** reaches the target angle **d5**, thus reducing a deviation of the angle of the imager **12** from the target angle **d5**.

### Components in Third Embodiment

[0140] FIG. **15** shows the subject **S** taking a selfie with an optical device **120** according to a third embodiment. The same components of the optical device **120** and the same processes in a method for controlling the optical device **120** as in the first and second embodiments are denoted with the same reference numerals and will not be described, or the figure numbers will not be referred to.

[0141] The method for controlling the optical device **120** differs from those according to the first and second embodiments in that the method uses the optical device **120** alone without using the communication terminal **100** or the communication terminal **110**. The optical device **120** differs from the optical device **10** in that the body case **11** is replaced with a case **122**, the case **122** includes the display panel **108**, and the controller **20** has the functions of the controller **102**. The other components are the same as in the first and second embodiments.

[0142] The optical device **120** is supported by a support **124**. The support **124** includes a rod **125** extendable in one direction and an attachment **126** fixed to the tip of the rod **125**. The case **122** includes a lower portion that is detachably attached to the attachment **126**. The subject S can take a selfie with the optical device **120** by extending the rod **125** in one direction with the case **122** attached to the attachment **126** and gripping an end portion of the rod **125**.

[0143] The selection frame FR that is quadrilateral appears on the display panel **108**. The selection frame FR can be moved to an intended position by touching the display panel **108** with a finger. A target in the selection frame FR is at the center of an image obtained by the imager **12**. In other words, the actuator **30** operates to change the angle of the imager **12** to cause a target specified with the selection frame FR to be at the center of an image.

[0144] FIG. **16** shows, in addition to the reference line M and the boundaries BL and UL, an imaginary line GA representing the actual changes in the angle of the imager **12**, a lower limit line DL1, and an upper limit line DL2. The lower limit line DL1 is defined in an area in which the deviation angle from the reference line M is greater than the deviation angle of the boundary BL. The upper limit line DL2 is defined in an area in which the deviation angle from the reference line M is greater than the deviation angle of the boundary UL. The range hatched with diagonal lines between the lower limit line DL1 and the upper limit line DL2 is defined as a limit range LR.

[0145] FIG. **16** shows the angle of the imager **12** exceeding the allowable range A1 and further exceeding the limit range LR (causing a deviation angle d03) during change of the orientation of the imager **12** (FIG. **15**) to the target angle d4 based on the reference line M. In FIG. **16**, the angle of the imager **12** is within the allowable range A1 along the reference line M until the angle of the imager **12** reaches an angle dF. However, in response to the optical device **120** (FIG. **15**) receiving hand-held camera shake at the angle dF, the angle of the imager **12** exceeds the limit range LR (lower limit line DL1) with the deviation angle d03.

[0146] The angle of the imager **12** exceeding the limit range LR is less likely to fall within the allowable range A1 when the actuator **30** (FIG. **15**) is operated. Thus, the optical device **120** prestores a program that ends the process of changing the angle of the imager **12** and the imaging performed by the imager **12** in response to the angle of the imager **12** exceeding the limit range LR.

### Effects of Components in Third Embodiment

[0147] A method for controlling the optical device **120** according to the third embodiment will now be described with reference to the flowchart shown in FIG. **17**. For the components of the optical device **120** (FIG. **15**), FIGS. **1** to **5** and FIGS. **15** and **16** are to be referred to, and the figure numbers are not individually referred to. The same effects as in the first embodiment will not be described.

[0148] The processes shown in FIG. **17** are performed by the CPU **22** that loads and executes a program (not shown) after reading the program from the memory **24**. For example, the program starts after the subject S operates the display panel **108** to set (performs face authentication on) the face SF of the subject S. The angle of the imager **12** is then changed to cause the set face SF to be at the center portion of the display panel **108**. The optical device **120** is pre-attached to the attachment **126**. The subject S has already started taking a selfie with the rod **125** extended.

[0149] The same steps as in the first and second embodiments are denoted with the same reference numerals and will not be described. More specifically, the flowchart in the third embodiment differs from the flowchart (FIG. **14**) in the second embodiment in that step S33 is set in place of step S32 and step S41 is added.

[0150] After step S30, the processing advances to step S33. In step S33, the CPU **104** defines the allowable range A1 and the limit range LR of vibrations based on the reference line M. The processing then advances to step S34. More specifically, the CPU **104** creates a table of combinations of the values of the tilt angle and the values of the pan angle, and then determines thresholds corresponding to the boundaries BL and UL and limit values corresponding to the lower limit line DL1 and the upper limit line DL2. The CPU **104** can thus apply the value of the tilt angle

and the value of the pan angle obtained by the detector **72** to the table to determine whether the angle of the imager **12** is within the allowable range **A1** and whether the angle of the imager **12** exceeds the limit range **LR**. Step **S33** is an example of setting a limit value greater than a threshold for the deviation angle in the intersecting direction.

[0151] In step **S40**, when the obtained deviation angle **d03** is greater than the threshold (Yes in **S40**), the processing advances to step **S41**. When the obtained deviation angle **d03** is less than or equal to the threshold (No in **S40**), the processing advances to step **S34**.

[0152] In step **S41**, the CPU **104** determines whether the deviation angle **d03** obtained in step **S40** is less than or equal to the limit values. More specifically, the CPU **104** defines an angular value corresponding to the difference between the reference line **M** and the lower limit line **DL1** and an angular value corresponding to the difference between the reference line **M** and the upper limit line **DL2** as the limit values, and compares the obtained deviation angle **d03** with the limit values. When the obtained deviation angle **d03** is less than or equal to the limit values (Yes in **S41**), the processing advances to step **S42**. When the obtained deviation angle **d03** is greater than the limit values (No in **S41**), the CPU **104** ends the program. Step **S41** is an example of stopping the operation of changing, with the actuator **30**, the angle of the imager **12** in response to the deviation angle **d03** being greater than the limit values.

[0153] When the angle of the imager **12** exceeds the limit range **LR** for the reference line **M**, the actuator **30** cannot correct the angle of the imager **12**. Driving the actuator **30** that cannot correct the angle of the imager **12** causes wasteful consumption of the battery power. The method for controlling the optical device **120** according to the third embodiment thus stops the operation of the actuator **30** when the deviation angle of the imager **12** in the intersecting direction is out of the limit range **LR** to reduce the likelihood of wasteful consumption of power by the actuator **30**.

#### Components in Fourth Embodiment

[0154] FIG. **18** shows the subject imaged by the imager **12** switched from the first subject **S1** to the second subject **S2** in an optical device **10** according to a fourth embodiment. The subject includes the first subject **S1** and the second subject **S2**. The same components of the optical device **10** and the communication terminal **100** as in the first embodiment are denoted with the same reference numerals and will not be described.

[0155] The optical device **10** is placed on the top of the desk **D**. The optical device **10** is supported on the support plate (not shown). For example, the first subject **S1** and the second subject **S2** are in the range of an angle of view **VB** of the imager **12**. In other words, the imager **12** is oriented toward the first subject **S1** and the second subject **S2** to allow the first subject **S1** and the second subject **S2** to be in the angle of view **VB**. The first subject **S1** is farther from the optical device **10**. The second subject **S2** is closer to the optical device **10**. The selection frame **FR** appears on the display panel **108**. The selection frame **FR** is specified by, for example, the second subject **S2** operating the communication terminal **100**.

[0156] FIG. **19** shows the reference line **M1** used when the orientation of the imager **12** (FIG. **18**) is changed from the initial angle **d0** to the target angle **d4** and a reference line **M4** used when the angle of the imager **12** is changed from the initial angle **d0** to a target angle **d6**. The reference lines **M1** and **M4** are each indicated by an arrow. For example, the reference line is switched from **M1** to **M4** from the time point at which the imager **12** reaches an angle **dH** on the reference line **M1** to the time point at which the imager **12** reaches an angle **dK** on the reference line **M4**. The reference for the angle of the imager **12** in switching the reference line from **M1** to **M4** is indicated by a reference line **M5**. The angle of the imager **12** is changed along the reference line **M4** and to the target angle **d6** after the reference line is switched from **M1** to **M5** at the angle **dH**. The reference line **M1** is an example of a first reference line. The reference line **M4** is an example of a second reference line.

[0157] The target angle **d4** is the angle of the imager **12** at which the imager **12** can capture an image of the first subject **S1**. The target angle **d6** is the angle of the imager **12** at which the imager

**12** can capture an image of the second subject **S2**. The target angle **d4** is represented by the tilt angle  $\theta B$  ( $^{\circ}$ ) and the pan angle  $\theta 2$  ( $^{\circ}$ ). The target angle **d6** is represented by the tilt angle  $\theta C$  ( $^{\circ}$ ) and a pan angle  $\theta 1$  ( $^{\circ}$ ).  $\theta A < \theta B < \theta C$  is satisfied, and  $\theta 0 < \theta 1 < \theta 2$  is satisfied. When a subject to be imaged is switched from the first subject **S1** to the second subject **S2**, both the tilt angle and the pan angle of the imager **12** are changed.

[0158] In FIG. **19**, an intersecting direction intersecting with the reference line **M1** is indicated by an arrow **Q3**. A direction intersecting with the reference line **M4** is indicated by an arrow **Q4**. In FIG. **19**, as described above, the specified subject is changed from the first subject **S1** to the second subject **S2** before the angle of the imager **12** reaches the target angle **d4**, and the angle of the imager **12** is changed to the target angle **d6**.

[0159] The allowable range at the initial angle **d0** is indicated by the circle **AR**. The allowable range at each of the target angles **d4** and **d6** is indicated by the circle **BR**. The lower limits for the reference line **M1** are indicated by a boundary **BL3**. The upper limits are indicated by a boundary **UL3**. The range between the boundary **BL3** and the boundary **UL3** is defined as an allowable range **A4** of the deviation angle. Similarly, the lower limits for the reference line **M4** are indicated by a boundary **BL4**. The upper limits are indicated by a boundary **UL4**. The range between the boundary **BL4** and the boundary **UL4** is defined as an allowable range **A5** of the deviation angle. The allowable ranges **A4** and **A5** are both smaller as the angle of the imager **12** is closer to the target angle **d4** or **d6**.

#### Effects of Components in Fourth Embodiment

[0160] A method for controlling the optical device **10** according to the fourth embodiment will now be described with reference to the flowchart shown in FIG. **20**. For the components of the optical device **10** and the communication terminal **100**, FIGS. **1** to **5** and FIGS. **18** and **19** are to be referred to, and the figure numbers are not individually referred to. The same effects as in the first embodiment will not be described.

[0161] The processes shown in FIG. **20** are performed by the CPU **22** that loads and executes a program (not shown) after reading the program from the memory **24**. For example, the program starts in response to the second subject **S2** operating the display panel **108** to set (perform face authentication on) a face **SF1** of the first subject **S1**. The angle of the imager **12** is then changed to cause the set face **SF1** to be at the center portion of the display panel **108**. In this example, the display panel **108** is operated to set a face **SF2** of the second subject **S2** during change of the angle of the imager **12** toward the first subject **S1**.

[0162] The same steps as in the first, second, and third embodiments are denoted with the same reference numerals and will not be described. More specifically, the flowchart in the fourth embodiment differs from the flowchart (FIG. **14**) in the second embodiment in that step **S37** is set in place of step **S44**.

[0163] Step **S20** is an example of specifying the first subject **S1** or the second subject **S2**. Steps **S30** and **S32** are examples of defining the reference line **M1** and setting a first threshold (allowable range **A4**) as the threshold in response to the first subject **S1** being specified. Step **S34** is an example of starting an operation of changing, with the actuator **30**, the angle of the imager **12** along the reference line **M1**.

[0164] Further, step **S34** is also an example of changing, with the actuator **30**, the angle of the imager **12** until the deviation angle in the intersecting direction falls below the first threshold in response to the deviation angle in the intersecting direction from the reference line **M1** exceeding the first threshold in changing, with the actuator **30**, the angle of the imager **12** along the reference line **M1**.

[0165] In step **S36**, the CPU **22** determines whether the angular difference is greater than  $1^{\circ}$ . When the angular difference is greater than  $1^{\circ}$  (Yes in **S36**), the processing advances to step **S37**. When the angular difference is less than or equal to  $1^{\circ}$  (No in **S36**), the CPU **22** determines that the imager **12** has reached the target angle **d4** and ends the program.

[0166] In step S37, the CPU 22 determines whether the target remains unchanged based on specification information from the selection frame FR. When the target remains unchanged (Yes in S37), the processing advances to step S38. When the target is changed (No in S37), or in other words, when the target is changed from the first subject S1 to the second subject S2, the processing advances to step S20. Step S37 and steps S20 to S32 after the target is changed are examples of setting the reference line M4 and a second threshold (allowable range A5) as the threshold in response to the second subject S2 being specified in changing, with the actuator 30, the angle of the imager 12 along the reference line M1. Steps S38 to S42 are the same as in the second embodiment.

[0167] Step S34 after the target is changed is an example of starting an operation of changing, with the actuator 30, the angle of the imager 12 along the reference line M4.

[0168] Further, steps S34 to S42 are examples of changing, with the actuator 30, the angle of the imager 12 until the deviation angle in the intersecting direction falls below the second threshold in response to the deviation angle in the intersecting direction from the reference line M4 exceeding the second threshold in changing, with the actuator 30, the angle of the imager 12 along the reference line M4.

[0169] As described above, the method for controlling the optical device 10 according to the fourth embodiment changes the reference line and the allowable range when the target is changed. As shown in FIGS. 18 and 19, during movement of the imager 12 with the actuator 30 to the target angle d6 to which the target angle has been changed, the angle of the imager 12 may exceed the boundary BL4 and reach an angle DL in response to vibrations received by the optical device 10. In this case, the actuator 30 corrects the angle (difference angle) of the imager 12 from the angle dL to an angle within the allowable range A5. This thus reduces image blur when the imaging target is changed from the first subject S1 to the second subject S2.

#### Modifications

[0170] The present invention is not limited to any of the first, second, third, and fourth embodiments described above, and can be modified variously without departing from the spirit and scope of the invention.

[0171] The number of magnetic poles of each of the first magnets 56 and the second magnets 58 may be different from the number of magnetic poles shown in FIG. 3.

[0172] In the actuator 30, the center for changing the angle in the pan direction and the center for changing the angle in the tilt direction may not be aligned with the reference point C. For example, a two-axis actuator displaceable in the X-direction or the Y-direction may be added, and the angles in the pan direction and the tilt direction may be changed after the actuator 30 is moved in the X-direction or the Y-direction. The angle of the imager 12 may not have the components in both the pan direction and the tilt direction, and may have a component in one of the pan direction or the tilt direction alone.

[0173] In the first, second, and fourth embodiments, the limit range LR may be set, and the operation of changing the angle of the imager 12 may be stopped in response to the angle of the imager 12 exceeding the limit range LR.

[0174] The acceleration AC of the actuator 30 may not be set smaller when the angle of the imager 12 is closer to the reference line M. The allowable ranges A1, A2, and A3 may not be set smaller as the angle of the imager 12 is closer to the target angle.

[0175] The target may not be specified with the selection frame FR operated on the display panel 108. For example, a list of target information (information about the faces of subjects) may be preset, the target may be selected from the list and registered, and face recognition may be performed automatically to specify the target.

[0176] The number of subjects may be three or more rather than one or two. In this case, the subject may be switched twice or more times, rather than once.

[0177] In the first, second, third, and fourth embodiments, the determination as to whether the

angle of the imager **12** is deviated (whether blur occurs) may be performed after a focus assembly (not shown) is operated to adjust the focal point.

[0178] The optical device **10** according to the first, second, and fourth embodiments may have the functions of the communication terminal **100** or **110**, and the optical device **10** alone may be used. In other words, the optical device **10** according to the first, second, and fourth embodiments may not be used with the communication terminal **100** or **110**.

[0179] The reference angle is an orientation of the optical axis for when the imager captures an image with the subject positioned at the center of the image. The deviation angle is the amount of deviation of the angle from the reference angle. The drive may not rotate the imager and may slide the imager in a set direction. With this structure, the orientation of the optical axis is substantially constant. To position the subject at the center of an image, an imaginary line connecting the center of the imager and the center of the subject is defined, and the imager is slid to align the optical axis direction with the direction of the imaginary line. In this case, a deviation between the direction of the imaginary line and the optical axis direction is a deviation (deviation angle) of the optical axis. In other words, in the structure in which the imager is slid as well, the deviation angle can be corrected with the imaginary line connecting the center of the imager and the center of the subject used as a reference line. The deviation angle in a direction intersecting with the reference line indicates vibrations (including hand-held camera shake) as disturbance. As described above, the control method according to one or more embodiments of the disclosure is also applicable to the structure in which the imager is slid.

[0180] The method according to one or more embodiments of the disclosure may have aspects described below.

[0181] (1) A method for controlling an optical device, the optical device including an imager to capture an image of at least one subject, a drive to change an angle of the imager, and an obtainer to obtain information about a deviation angle from a reference angle of the imager, the method comprising: [0182] specifying the at least one subject; [0183] defining, in response to the at least one subject being specified, an imaginary line connecting a center of the imager and a center of the at least one subject as a reference line representing the reference angle; [0184] setting at least one threshold of the deviation angle from the reference angle of the imager in an intersecting direction intersecting with the reference line; [0185] starting an operation of changing, with the drive, the angle of the imager along the reference line; [0186] obtaining, with the obtainer, the information about the deviation angle in the intersecting direction; and [0187] changing, with the drive, the angle of the imager until the deviation angle in the intersecting direction falls below the at least one threshold in response to the deviation angle in the intersecting direction exceeding the at least one threshold in changing, with the drive, the angle of the imager along the reference line, and continuing the operation of changing, with the drive, the angle of the imager along the reference line after the deviation angle in the intersecting direction falls below the at least one threshold.

[0188] (2) The method according to (1), further comprising: [0189] specifying a first subject or a second subject, the first subject and the second subject being included in the at least one subject; [0190] defining a first reference line as the reference line and setting a first threshold as the at least one threshold in response to the first subject being specified; [0191] starting an operation of changing, with the drive, the angle of the imager along the first reference line; [0192] changing, with the drive, the angle of the imager until the deviation angle in the intersecting direction falls below the first threshold in response to the deviation angle in the intersecting direction from the first reference line exceeding the first threshold in changing, with the drive, the angle of the imager along the first reference line; [0193] defining a second reference line as the reference line and setting a second threshold as the at least one threshold in response to the second subject being specified in changing, with the drive, the angle of the imager along the first reference line; [0194] starting an operation of changing, with the drive, the angle of the imager along the second reference line; and [0195] changing, with the drive, the angle of the imager until the deviation angle in the

intersecting direction falls below the second threshold in response to the deviation angle in the intersecting direction from the second reference line exceeding the second threshold in changing, with the drive, the angle of the imager along the second reference line.

[0196] (3) The method according to (1), wherein [0197] the optical device operates by transmitting and receiving information to and from a communication terminal, [0198] the communication terminal includes [0199] an identifier to identify the at least one subject, and [0200] a movement detector to detect movement of the identified at least one subject from a first position to a second position, and [0201] the method comprises identifying, with the identifier, the at least one subject in response to the at least one subject being specified, [0202] defining, in response to the at least one subject being at the first position, an imaginary line connecting the center of the imager and the first position as the reference line before the movement, [0203] defining, in response to the movement detector detecting the movement of the at least one subject to the second position, an imaginary line connecting the center of the imager and the second position as a new reference line, [0204] setting, in response to the at least one subject being at the second position, a new threshold corresponding to the new reference line, [0205] changing the angle of the imager along the new reference line, and [0206] changing, with the drive, the angle of the imager until the deviation angle falls below the new threshold in response to the deviation angle exceeding the new threshold in changing the angle of the imager with the drive, and continuing an operation of changing, with the drive, the angle of the imager along the new reference line after the deviation angle falls below the new threshold.

[0207] (4) The method according to any one of (1) to (3), wherein [0208] the at least one threshold includes a subject threshold and an imager threshold set for an area closer to the imager than the subject threshold, and [0209] the subject threshold is less than the imager threshold.

[0210] (5) The method according to any one of (1) to (4), wherein [0211] the drive changing the angle of the imager has a lower acceleration when the angle of the imager is closer to the reference line.

[0212] (6) The method according to any one of (1) to (5), further comprising: [0213] setting, for the deviation angle in the intersecting direction, a limit value greater than the at least one threshold; and [0214] stopping the operation of changing, with the drive, the angle of the imager in response to the deviation angle being greater than the limit value.

[0215] (7) The method according to any one of (1) to (6), wherein [0216] the angle of the imager has components in a pan direction and a tilt direction perpendicular to each other, and [0217] the drive changes the angle of the imager in the pan direction and the tilt direction.

## Claims

1. A method for controlling an optical device, the optical device including an imager to capture an image of at least one subject, a drive to change an angle of the imager, and an obtainer to obtain information about a deviation angle from a reference angle of the imager, the method comprising: specifying the at least one subject; defining, in response to the at least one subject being specified, an imaginary line connecting a center of the imager and a center of the at least one subject as a reference line representing the reference angle; setting at least one threshold of the deviation angle from the reference angle of the imager in an intersecting direction intersecting with the reference line; starting an operation of changing, with the drive, the angle of the imager along the reference line; obtaining, with the obtainer, the information about the deviation angle in the intersecting direction; and changing, with the drive, the angle of the imager until the deviation angle in the intersecting direction falls below the at least one threshold in response to the deviation angle in the intersecting direction exceeding the at least one threshold in changing, with the drive, the angle of the imager along the reference line, and continuing the operation of changing, with the drive, the angle of the imager along the reference line after the deviation angle in the intersecting direction

falls below the at least one threshold.

2. The method according to claim 1, further comprising: specifying a first subject or a second subject, the first subject and the second subject being included in the at least one subject; defining a first reference line as the reference line and setting a first threshold as the at least one threshold in response to the first subject being specified; starting an operation of changing, with the drive, the angle of the imager along the first reference line; changing, with the drive, the angle of the imager until the deviation angle in the intersecting direction falls below the first threshold in response to the deviation angle in the intersecting direction from the first reference line exceeding the first threshold in changing, with the drive, the angle of the imager along the first reference line; defining a second reference line as the reference line and setting a second threshold as the at least one threshold in response to the second subject being specified in changing, with the drive, the angle of the imager along the first reference line; starting an operation of changing, with the drive, the angle of the imager along the second reference line; and changing, with the drive, the angle of the imager until the deviation angle in the intersecting direction falls below the second threshold in response to the deviation angle in the intersecting direction from the second reference line exceeding the second threshold in changing, with the drive, the angle of the imager along the second reference line.

3. The method according to claim 1, wherein the optical device operates by transmitting and receiving information to and from a communication terminal, the communication terminal includes an identifier to identify the at least one subject, and a movement detector to detect movement of the identified at least one subject from a first position to a second position, and the method comprises identifying, with the identifier, the at least one subject in response to the at least one subject being specified, defining, in response to the at least one subject being at the first position, an imaginary line connecting the center of the imager and the first position as the reference line before the movement, defining, in response to the movement detector detecting the movement of the at least one subject to the second position, an imaginary line connecting the center of the imager and the second position as a new reference line, setting, in response to the at least one subject being at the second position, a new threshold corresponding to the new reference line, changing the angle of the imager along the new reference line, and changing, with the drive, the angle of the imager until the deviation angle falls below the new threshold in response to the deviation angle exceeding the new threshold in changing the angle of the imager with the drive, and continuing an operation of changing, with the drive, the angle of the imager along the new reference line after the deviation angle falls below the new threshold.

4. The method according to claim 3, wherein the at least one threshold includes a subject threshold and an imager threshold set for an area closer to the imager than the subject threshold, and the subject threshold is less than the imager threshold.

5. The method according to claim 4, wherein the drive changing the angle of the imager has a lower acceleration when the angle of the imager is closer to the reference line.

6. The method according to claim 1, further comprising: setting, for the deviation angle in the intersecting direction, a limit value greater than the at least one threshold; and stopping the operation of changing, with the drive, the angle of the imager in response to the deviation angle being greater than the limit value.

7. The method according to claim 1, wherein the angle of the imager has components in a pan direction and a tilt direction perpendicular to each other, and the drive changes the angle of the imager in the pan direction and the tilt direction.

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